

HE2001 Microeconomics II

Tutorial 10

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Overview

This tutorial covers:

- plurality voting and Borda count as social decision mechanisms;
- completeness, transitivity, the Pareto condition, and independence of irrelevant alternatives;
- strategic incentives under rank-based voting rules;
- utility possibility frontiers and their slopes;
- utilitarian, weighted-utilitarian, Nietzschean, Rawlsian, and Cobb–Douglas social welfare functions;
- graphical and analytical characterisation of socially optimal utility allocations.

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Question 1: Plurality Voting

Consider Plurality Voting as discussed in the lecture.¹ Suppose that there are a finite number of alternatives and that everyone has rational and strict preferences.

Problem

- (a) Will the plurality voting social decision mechanism always result in a complete and transitive social preference ordering? Explain.
- (b) Does plurality voting satisfy the pareto condition? Explain.
- (c) Consider the case of 3 alternatives and a society with 9 individuals with the following preference list:
- 4 individuals have $A \succ_i B \succ_i C$
 - 3 individuals have $C \succ_i B \succ_i A$
 - 2 individuals have $B \succ_i A \succ_i C$

What is the social ranking according to the plurality voting?

- (d) Now, suppose we swap the middle 3 individuals' preferences over B and C such that the preference list becomes:
- 4 individuals have $A \succ_i B \succ_i C$
 - 3 individuals have $B \succ_i C \succ_i A$
 - 2 individuals have $B \succ_i A \succ_i C$

What is the social ranking according to plurality voting?

- (e) Does the plurality voting social decision mechanism satisfy the requirement of independence of irrelevant alternatives? Hint: use parts c) and d).

¹Suppose that there are n alternatives. Given the n alternatives, pick the alternative that is the top choice of the largest number of individuals and place it at the top of the social preference ordering. Ties are broken randomly. Then, remove that alternative from every individual's preference ordering, keeping the same order for the $n-1$ remaining alternatives. (For example, if $a \succ_i b \succ_i c$, and we are removing alternative b which was the top, then it becomes $a \succ_i c$.) Repeat the above process on the remaining alternatives until only one remains which is the bottom of the social preference ordering.

Solution

Method 1: Recursive winner-elimination method

- (a) Yes. At each stage of the plurality procedure, one of the remaining alternatives is selected and placed next in the social ordering.

There are only finitely many alternatives. Therefore the procedure cannot continue forever: after one alternative is chosen, the number of remaining alternatives falls by one, and after finitely many steps exactly one alternative remains.

Hence the procedure always produces an ordered list of all alternatives, say

$$x_1 \succ x_2 \succ \cdots \succ x_n.$$

Because every pair of alternatives appears somewhere in this list, the social ordering is complete. Because the ranking is literally a single ordered list, it is also transitive: whenever

$$x_i \succ x_j \quad \text{and} \quad x_j \succ x_k,$$

we automatically have

$$x_i \succ x_k.$$

Therefore plurality voting always yields

a complete and transitive social preference ordering.

- (b) Yes. Suppose every individual strictly prefers some alternative X to another alternative Y .

Consider any round of the plurality procedure in which both X and Y are still present. Since every person ranks X above Y , it is impossible for Y to be anyone's top-ranked remaining option unless X has already been removed earlier.

But that earlier removal would mean that X had already been placed socially above Y . Therefore there is no possible stage at which Y can be placed above X while X remains below it.

Hence plurality voting respects unanimous strict preferences:

If everyone prefers X to Y , then society ranks X above Y .

So the Pareto condition is satisfied.

- (c) In the first round, the first-place vote totals are:

$$A : 4, \quad B : 2, \quad C : 3.$$

Hence A has the largest number of first-place votes, so A is placed first socially.

Remove A from every ranking. The remaining rankings become:

- 4 individuals: $B \succ_i C$

- 3 individuals: $C \succ_i B$
- 2 individuals: $B \succ_i C$

So the new first-place totals are:

$$B : 6, \quad C : 3.$$

Hence B is placed second and C last.

Therefore the plurality-voting social ranking is

$$\boxed{A \succ B \succ C.}$$

(d) Now the first-round first-place vote totals are:

$$A : 4, \quad B : 5, \quad C : 0.$$

Hence B is placed first socially.

Remove B from every ranking. The remaining rankings become:

- 4 individuals: $A \succ_i C$
- 3 individuals: $C \succ_i A$
- 2 individuals: $A \succ_i C$

So the new first-place totals are:

$$A : 6, \quad C : 3.$$

Hence A is placed second and C last.

Therefore the social ranking is

$$\boxed{B \succ A \succ C.}$$

(e) Independence of irrelevant alternatives requires that if every individual's ranking between A and B stays unchanged, then the social ranking between A and B should also stay unchanged.

Compare parts (c) and (d). In both profiles:

- the 4 individuals still rank A above B ;
- the 3 middle individuals still rank B above A ;
- the 2 remaining individuals still rank B above A .

So nobody changed their relative ranking of A and B .

However, the social ranking between A and B changed:

$$A \succ B \quad \text{in part (c),} \quad B \succ A \quad \text{in part (d).}$$

Hence plurality voting does not satisfy independence of irrelevant alternatives.

Therefore

$$\boxed{\text{Plurality voting violates IIA.}}$$

Method 2: Top-count tables and contradiction argument

- (a) The plurality rule sequentially assigns first place, then second place, and so on, until every alternative has been assigned a position. This necessarily generates one ranked list of all alternatives. Any such ranked list is complete and transitive. Hence

plurality voting always gives a complete and transitive ordering.

- (b) Suppose everybody ranks X above Y . Then whenever both X and Y remain under consideration, Y cannot receive more top rankings than X , because Y is never above X in any person's list. So Y cannot be placed socially above X . Hence the Pareto condition holds:

$X \succ Y$ socially whenever all individuals have $X \succ_i Y$.

- (c) The plurality table for the original profile is

Round 1	A	B	C
Top votes	4	2	3

so A is first.

After removing A ,

Round 2	B	C
Top votes	6	3

so B is second and C third. Thus

$A \succ B \succ C$.

- (d) The plurality table for the modified profile is

Round 1	A	B	C
Top votes	4	5	0

so B is first.

After removing B ,

Round 2	A	C
Top votes	6	3

so A is second and C third. Thus

$B \succ A \succ C$.

- (e) In parts (c) and (d), all individuals keep the same relative ordering of A and B , but the social ordering between A and B reverses. Therefore plurality voting depends on changes involving the third option C , which is exactly what IIA forbids. Hence

plurality voting fails IIA.

Question 2: Borda Count

Consider the Borda Count as discussed in the lecture. Suppose that there are a finite number of alternatives and that everyone has rational and strict preferences.

Problem

- (a) Will the Borda Count social decision mechanism always result in a complete and transitive social preference ordering? Explain.
- (b) Does the Borda Count social decision mechanism defined in this way satisfy the pareto condition? Explain.
- (c) Suppose that there are two voters and three candidates, x , y , and z . Suppose that Voter 1 ranks the candidates: x first, z second, and y third. Suppose that Voter 2 ranks the candidates: y first, x second, and z third. What is the Borda count for x , y and z ?
- (d) Now suppose that it is discovered that candidate z once lifted a beagle by the ears. Voter 1, who has rather large ears himself, is appalled and changes his ranking to x first, y second, z third. Voter 2, who picks up his own children by the ears, is favorably impressed and changes his ranking to y first, z second, x third. What is the Borda count for x , y and z ?
- (e) Does the Borda Count social decision mechanism satisfy independence of irrelevant alternatives? Hint: use parts d) and e).
- (f) Do you think that with Borda Voting, voters always have an incentive to truthfully report their full preference rankings? (Hint: provide an example where a voter changes his reported ranking and benefits from it)

Solution

Method 1: Score-summation method

- (a) Yes. Under the Borda Count, every alternative receives a total score obtained by summing the points assigned by all voters.

Because the number of voters and alternatives is finite, every alternative has a well-defined total score. Comparing total scores yields a social ordering: the alternative with the smaller total score is socially preferred, and equal total scores imply social indifference under this version of the rule.

Since ordinary numerical comparison is complete and transitive, the induced social ordering is also complete and transitive. Therefore

the Borda Count always produces a complete and transitive social preference ordering.

- (b) Yes. Suppose every voter ranks x above y .

Under the Borda Count, a higher-ranked alternative receives a strictly better score from each individual than a lower-ranked alternative. Therefore each voter assigns fewer points to x than to y . Summing across all voters preserves this strict inequality, so the total score of x is smaller than the total score of y .

Since lower total score is socially better, society ranks x above y . Hence the Borda Count satisfies the Pareto condition:

If every voter ranks x above y , then society ranks x above y .

- (c) With three candidates, assign 1 point to first place, 2 points to second place, and 3 points to third place.

Voter 1 has

$$x \succ z \succ y,$$

so contributes

$$x : 1, \quad z : 2, \quad y : 3.$$

Voter 2 has

$$y \succ x \succ z,$$

so contributes

$$y : 1, \quad x : 2, \quad z : 3.$$

Summing these scores:

$$x : 1 + 2 = 3, \quad y : 3 + 1 = 4, \quad z : 2 + 3 = 5.$$

Therefore the Borda counts are

$$x : 3, \quad y : 4, \quad z : 5.$$

Since lower score is better, the social ranking is

$$x \succ y \succ z.$$

(d) Under the revised rankings,

$$\text{Voter 1: } x \succ y \succ z, \quad \text{Voter 2: } y \succ z \succ x.$$

So Voter 1 contributes

$$x : 1, \quad y : 2, \quad z : 3,$$

and Voter 2 contributes

$$y : 1, \quad z : 2, \quad x : 3.$$

Summing:

$$x : 1 + 3 = 4, \quad y : 2 + 1 = 3, \quad z : 3 + 2 = 5.$$

Therefore the Borda counts are

$$x : 4, \quad y : 3, \quad z : 5.$$

Hence the social ranking is

$$y \succ x \succ z.$$

(e) No. Compare parts (c) and (d). Nobody changed their mind about the relative ordering of x and y :

- Voter 1 ranks x above y in both cases.
- Voter 2 ranks y above x in both cases.

So each voter's preferences between x and y are unchanged.

However, the social ranking between x and y changes:

$$x \succ y \quad \text{in part (c),} \quad y \succ x \quad \text{in part (d).}$$

The change arose only because the position of the third candidate z changed. Therefore Borda Count violates independence of irrelevant alternatives:

$$\text{Borda Count does not satisfy IIA.}$$

(f) No. Voters do not always have an incentive to report truthfully.

Consider the following true preferences of three voters over four alternatives X, Y, Z, W :

Rank	Bill	Bertha	Bob
1	X	Y	Z
2	Y	Z	X
3	Z	X	Y
4	W	W	W

With four alternatives, assign 1 point to first place, 2 to second, 3 to third, and 4 to fourth. Under truthful reporting the totals are

$$X : 1 + 3 + 2 = 6, \quad Y : 2 + 1 + 3 = 6, \quad Z : 3 + 2 + 1 = 6, \quad W : 4 + 4 + 4 = 12.$$

So X, Y, Z tie for the best score. If ties are broken randomly, Bertha faces an equal-chance outcome among X, Y, Z .

Now suppose Bertha strategically reports

$$Y \succ W \succ Z \succ X$$

instead of her true ranking

$$Y \succ Z \succ X \succ W.$$

Then the reported profile becomes

Rank	Bill	Bertha	Bob
1	X	Y	Z
2	Y	W	X
3	Z	Z	Y
4	W	X	W

and the totals become

$$X : 1 + 4 + 2 = 7, \quad Y : 2 + 1 + 3 = 6, \quad Z : 3 + 3 + 1 = 7, \quad W : 4 + 2 + 4 = 10.$$

So Y becomes the unique social winner.

Since Bertha's true ranking has Y as her top choice, she strictly prefers this manipulated outcome to the truthful tie among X, Y, Z . Hence she benefits from misreporting.

Therefore

Borda voting is not strategy-proof; voters may gain from lying.

Method 2: Rank-order proof and explicit manipulation example

- (a) The Borda rule converts each preference profile into a vector of total scores. Ordering alternatives by their total scores gives a social ranking. Because real numbers are completely and transitively ordered, the resulting social preference relation is complete and transitive. Thus

the Borda Count always yields a complete and transitive ordering.

- (b) If every voter ranks x above y , then x receives a better rank than y from every voter, hence fewer Borda points from every voter, and therefore fewer total points overall. So society ranks x above y . Hence

the Borda Count satisfies the Pareto condition.

- (c) Under the original profile, the score table is

	x	y	z
Voter 1	1	3	2
Voter 2	2	1	3
Total	3	4	5

so

$$x \succ y \succ z.$$

(d) Under the revised profile, the score table is

	x	y	z
Voter 1	1	2	3
Voter 2	3	1	2
Total	4	3	5

so

$$y \succ x \succ z.$$

(e) The relative individual ranking between x and y is unchanged between parts (c) and (d), yet the social ranking flips from $x \succ y$ to $y \succ x$. Therefore the rule depends on changes in the position of z , which means

Borda Count fails IIA.

(f) The strategic example in Method 1 proves that truthful revelation need not be optimal. The key mechanism is that under Borda Count, changing the reported ranking of lower alternatives changes their scores and can alter the winner, even when the voter's favourite remains at the top. Thus

voters may benefit from misreporting under Borda voting.

Question 3: Norton, Ralph, and the Utility Possibility Frontier

Norton and Ralph have a utility possibility frontier that is given by the following equation,

$$U_R + U_N^2 = 100$$

(where R and N signify Ralph and Norton respectively).

Problem

- (a) If we set Norton's utility to zero, what is the highest possible utility Ralph can achieve?
If we set Ralph's utility to zero, what is the best Norton can do?
- (b) Plot the utility possibility frontier on a graph.
- (c) Derive an equation for the slope of the above utility possibility curve.
- (d) Suppose that a social planner who is deciding on the allocation of Ralph and Norton has a utilitarian social welfare function. What is the utility of Ralph and Norton in the social planner's ideal allocation?
- (e) Both Ralph and Norton believe that the ideal allocation is given by maximizing an appropriate social welfare function. Ralph thinks that $U_R = 75, U_N = 5$ is the best distribution of welfare, and presents the maximization solution to a weighted-sum-of-the-utilities social welfare function that confirms this observation. What was Ralph's social welfare function?
(Hint: What is the slope of Ralph's social welfare function?)
- (f) Norton, on the other hand, believes that $U_R = 19, U_N = 9$ is the best distribution. What was Norton's social welfare function?

Solution

Method 1: Frontier substitution and tangency conditions

(a) The utility possibility frontier is

$$U_R + U_N^2 = 100.$$

If $U_N = 0$, then

$$U_R = 100 - 0^2 = 100.$$

Hence the highest utility Ralph can achieve is

$$U_R = 100.$$

If $U_R = 0$, then

$$U_N^2 = 100.$$

Taking the nonnegative value relevant for utility,

$$U_N = 10.$$

Hence the best Norton can do is

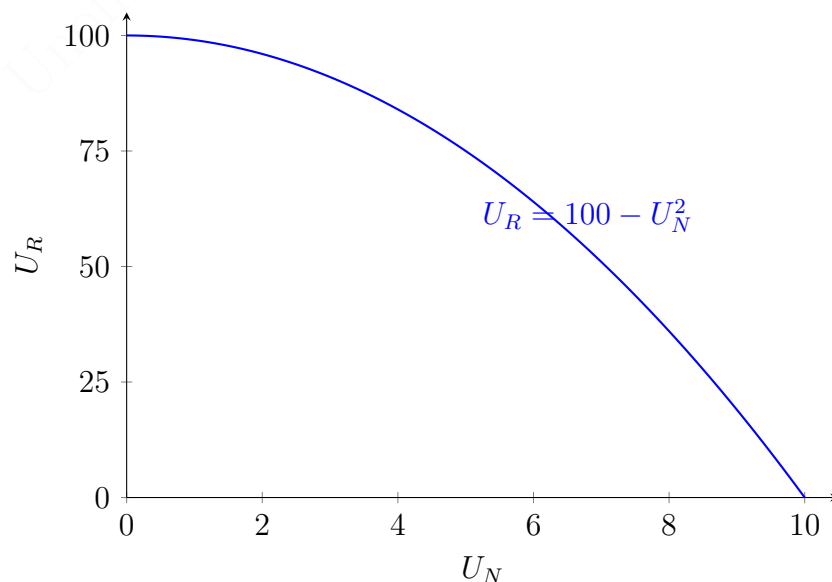
$$U_N = 10.$$

(b) Rewrite the frontier as

$$U_R = 100 - U_N^2.$$

Thus the frontier is a downward-sloping concave parabola in the (U_N, U_R) plane, running from $(0, 100)$ to $(10, 0)$.

A correct graph is shown below.



(c) Differentiate

$$U_R = 100 - U_N^2.$$

Therefore the slope of the utility possibility frontier is

$$\frac{dU_R}{dU_N} = -2U_N.$$

Hence

$$\boxed{\frac{dU_R}{dU_N} = -2U_N.}$$

(d) A utilitarian planner maximizes

$$W(U_R, U_N) = U_R + U_N$$

subject to

$$U_R + U_N^2 = 100.$$

Substitute the frontier into welfare:

$$W(U_N) = 100 - U_N^2 + U_N.$$

Differentiate:

$$W'(U_N) = -2U_N + 1.$$

Set equal to zero:

$$-2U_N + 1 = 0 \quad \implies \quad U_N = \frac{1}{2}.$$

Then

$$U_R = 100 - \left(\frac{1}{2}\right)^2 = 100 - \frac{1}{4} = \frac{399}{4}.$$

Therefore the utilitarian optimum is

$$\boxed{U_N = \frac{1}{2}, \quad U_R = \frac{399}{4} = 99.75.}$$

(e) Let Ralph's weighted-utilitarian social welfare function be

$$W(U_R, U_N) = aU_R + bU_N, \quad a, b > 0.$$

Its isowelfare lines satisfy

$$aU_R + bU_N = \bar{W},$$

so their slope is

$$\frac{dU_R}{dU_N} = -\frac{b}{a}.$$

At Ralph's desired optimum,

$$(U_R, U_N) = (75, 5).$$

The slope of the frontier there is

$$\frac{dU_R}{dU_N} = -2U_N = -10.$$

Tangency therefore requires

$$-\frac{b}{a} = -10 \quad \implies \quad \frac{b}{a} = 10.$$

Hence any positive weights with ratio $b : a = 10 : 1$ work. One example is

$$\boxed{W(U_R, U_N) = U_R + 10U_N.}$$

(f) At Norton's preferred allocation,

$$(U_R, U_N) = (19, 9).$$

The slope of the frontier there is

$$\frac{dU_R}{dU_N} = -2U_N = -18.$$

For a weighted-utilitarian function

$$W(U_R, U_N) = aU_R + bU_N,$$

tangency requires

$$-\frac{b}{a} = -18 \quad \implies \quad \frac{b}{a} = 18.$$

Hence any weights with ratio $18 : 1$ work. One example is

$$\boxed{W(U_R, U_N) = U_R + 18U_N.}$$

Method 2: Lagrangian and direct weighted-sum verification

- (a) Setting one person's utility to zero directly gives the frontier intercepts:

$$(U_N, U_R) = (0, 100) \quad \text{and} \quad (10, 0).$$

Therefore

$$\boxed{\text{Ralph's maximum is 100, Norton's maximum is 10.}}$$

- (b) The frontier is the graph of $U_R = 100 - U_N^2$, so it is a parabola opening downward in the (U_N, U_R) plane, beginning at $(0, 100)$ and ending at $(10, 0)$.
- (c) Implicit differentiation of

$$U_R + U_N^2 = 100$$

gives

$$dU_R + 2U_N dU_N = 0 \quad \implies \quad \frac{dU_R}{dU_N} = -2U_N.$$

So

$$\boxed{\frac{dU_R}{dU_N} = -2U_N.}$$

- (d) Use the Lagrangian

$$\mathcal{L} = U_R + U_N + \lambda(100 - U_R - U_N^2).$$

The first-order conditions are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial U_R} &= 1 - \lambda = 0, \\ \frac{\partial \mathcal{L}}{\partial U_N} &= 1 - 2\lambda U_N = 0. \end{aligned}$$

From the first equation, $\lambda = 1$. Substitute into the second:

$$1 - 2U_N = 0 \quad \implies \quad U_N = \frac{1}{2}.$$

Then the constraint gives

$$U_R = 100 - \frac{1}{4} = \frac{399}{4}.$$

Hence

$$\boxed{U_N = \frac{1}{2}, \quad U_R = \frac{399}{4}.}$$

- (e) Verify directly that a weighted-sum objective is maximized at $(75, 5)$ when the weight ratio is 10 : 1.

Let

$$W(U_R, U_N) = U_R + 10U_N.$$

On the frontier,

$$W(U_N) = 100 - U_N^2 + 10U_N.$$

Differentiate:

$$W'(U_N) = -2U_N + 10.$$

Set equal to zero:

$$-2U_N + 10 = 0 \quad \implies \quad U_N = 5.$$

Then

$$U_R = 100 - 25 = 75.$$

So indeed

$$\boxed{W(U_R, U_N) = U_R + 10U_N}$$

supports Ralph's preferred point.

(f) Similarly, take

$$W(U_R, U_N) = U_R + 18U_N.$$

On the frontier,

$$W(U_N) = 100 - U_N^2 + 18U_N.$$

Differentiating,

$$W'(U_N) = -2U_N + 18.$$

The maximizer satisfies

$$U_N = 9,$$

and then

$$U_R = 100 - 81 = 19.$$

Hence Norton's preferred point is supported by

$$\boxed{W(U_R, U_N) = U_R + 18U_N.}$$

Question 4: Social Welfare on a Linear Utility Possibility Frontier

Suppose the utility possibility frontier for two individuals is given by

$$U_A + 2U_B = 200.$$

Problem

- (a) Plot the utility possibility frontier.
- (b) In order to maximize a “Nietzschean social welfare function,” $W(U_A, U_B) = \max\{U_A, U_B\}$, what values of U_A, U_B would one set? Draw a few isowelfare lines and the social optimal point.
- (c) If instead we use a Rawlsian criterion, $W(U_A, U_B) = \min\{U_A, U_B\}$, what values of U_A, U_B would maximize the social welfare function? Draw a few isowelfare lines and the social optimal point.
- (d) Suppose that social welfare is given by $W(U_A, U_B) = U_A^{1/2}U_B^{1/2}$, what values of U_A, U_B would maximize the social welfare function? Draw a few isowelfare lines and the social optimal point.

Solution

Method 1: Graphical optimisation and frontier substitution

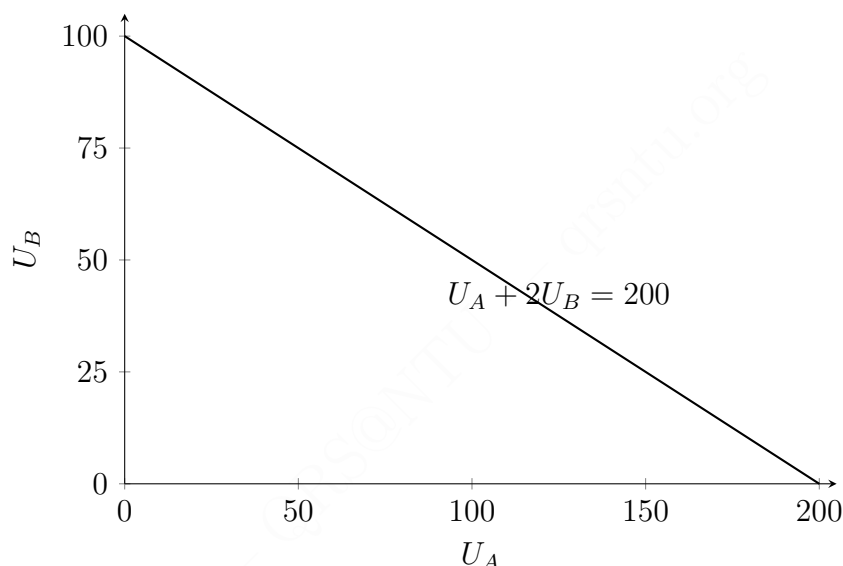
(a) Rewrite the frontier as

$$U_B = 100 - \frac{1}{2}U_A.$$

Hence the utility possibility frontier is a straight line with intercepts

$$(U_A, U_B) = (200, 0) \quad \text{and} \quad (0, 100).$$

A correct graph is shown below.



(b) The Nietzschean welfare function is

$$W(U_A, U_B) = \max\{U_A, U_B\}.$$

To maximize the maximum of the two utilities, we want the highest possible value attained by either person.

On the frontier,

$$U_A \leq 200, \quad U_B \leq 100.$$

The largest possible value of the maximum is therefore 200, achieved when

$$U_A = 200, \quad U_B = 0.$$

Thus the Nietzschean optimum is

$$\boxed{(U_A, U_B) = (200, 0)}.$$

The isowelfare lines for $\max\{U_A, U_B\} = \bar{W}$ are inverted-L shapes consisting of

$$U_A = \bar{W}, U_B \leq \bar{W} \quad \text{and} \quad U_B = \bar{W}, U_A \leq \bar{W}.$$

(c) The Rawlsian welfare function is

$$W(U_A, U_B) = \min\{U_A, U_B\}.$$

To maximize the minimum utility, we equalize the two utilities if possible. So impose

$$U_A = U_B.$$

Substitute into the frontier:

$$U_A + 2U_A = 200 \quad \implies \quad 3U_A = 200.$$

Hence

$$U_A = \frac{200}{3}, \quad U_B = \frac{200}{3}.$$

Therefore the Rawlsian optimum is

$$(U_A, U_B) = \left(\frac{200}{3}, \frac{200}{3}\right).$$

The isowelfare lines for $\min\{U_A, U_B\} = \bar{W}$ are L-shapes with kink at $(\bar{W}, \bar{W})$.

(d) Now social welfare is

$$W(U_A, U_B) = U_A^{1/2} U_B^{1/2}.$$

Since the square root is increasing, maximizing W is equivalent to maximizing

$$U_A U_B.$$

Use the frontier to substitute for U_B :

$$U_B = 100 - \frac{1}{2}U_A.$$

So we maximize

$$f(U_A) = U_A \left(100 - \frac{1}{2}U_A\right) = 100U_A - \frac{1}{2}U_A^2.$$

Differentiate:

$$f'(U_A) = 100 - U_A.$$

Setting $f'(U_A) = 0$ gives

$$U_A = 100.$$

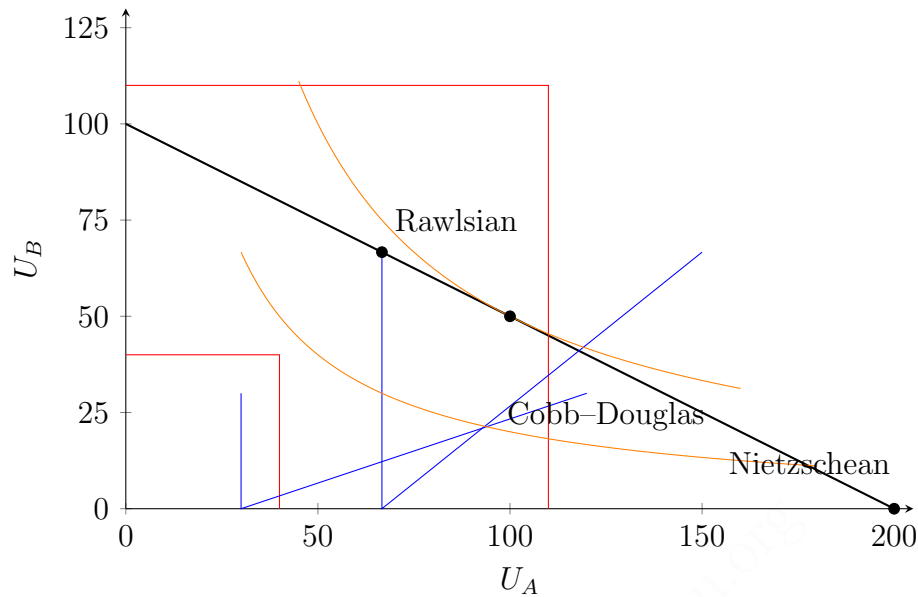
Then

$$U_B = 100 - \frac{1}{2}(100) = 50.$$

Therefore the social optimum is

$$(U_A, U_B) = (100, 50).$$

A combined sketch for parts (b)–(d) is shown below.



Method 2: Slope matching and direct objective analysis

(a) From

$$U_A + 2U_B = 200,$$

implicit differentiation gives

$$1 + 2 \frac{dU_B}{dU_A} = 0 \quad \Rightarrow \quad \frac{dU_B}{dU_A} = -\frac{1}{2}.$$

So the frontier is a straight line of slope $-1/2$ with intercepts $(200, 0)$ and $(0, 100)$.

(b) For $W(U_A, U_B) = \max\{U_A, U_B\}$, the greatest possible welfare is obtained by making the larger of the two utilities as large as possible. Along the frontier the maximum achievable utility for person A is 200, whereas for person B it is only 100. Therefore the welfare-maximizing point is

$$(U_A, U_B) = (200, 0).$$

(c) For $W(U_A, U_B) = \min\{U_A, U_B\}$, the welfare level equals the smaller of the two utilities. The maximum of the minimum occurs where the two utilities are equal, because otherwise one could raise the smaller one without lowering the larger below it until equality is reached. Thus impose

$$U_A = U_B.$$

Combining with the frontier gives

$$3U_A = 200, \quad U_A = U_B = \frac{200}{3}.$$

Hence

$$(U_A, U_B) = \left(\frac{200}{3}, \frac{200}{3} \right).$$

(d) For

$$W(U_A, U_B) = U_A^{1/2} U_B^{1/2},$$

an isowelfare curve satisfies

$$U_A^{1/2} U_B^{1/2} = \bar{W}.$$

Taking logs or differentiating directly,

$$\frac{dU_B}{dU_A} = -\frac{U_B}{U_A}$$

along an isowelfare curve.

At an interior optimum, the slope of the isowelfare curve equals the slope of the frontier:

$$-\frac{U_B}{U_A} = -\frac{1}{2}.$$

Hence

$$U_A = 2U_B.$$

Substitute into the frontier:

$$2U_B + 2U_B = 200 \quad \implies \quad 4U_B = 200 \quad \implies \quad U_B = 50.$$

Therefore

$$U_A = 100.$$

So the optimum is

$$\boxed{(U_A, U_B) = (100, 50)}.$$

Question 1

Solution

There are three policies X, Y, Z , and three preference types:

Young: $Z \succ Y \succ X$, Middle Aged: $X \succ Z \succ Y$, Old: $Y \succ X \succ Z$.

The numbers of voters are:

300 young, 300 middle aged, 400 old.

Under *standard Borda voting* with three alternatives, each voter awards:

1st place = 2 points, 2nd place = 1 point, 3rd place = 0 points.

Under the *modified Borda voting* rule, the same ranking scores apply, except:

- young voters' points are multiplied by 2,
 - middle-aged voters' points are multiplied by 1.5,
 - old voters' points remain unchanged.
- (a) In general, the modified Borda rule gives *more weight* to the preferences of young and middle-aged consumers, relative to old consumers.

Under standard Borda, each person's ranking contributes equally. Under the modified rule:

Young weight = 2, Middle-aged weight = 1.5, Old weight = 1.

So the social ranking will tend to shift toward alternatives that are ranked highly by the young and middle-aged groups, and away from alternatives that are mainly supported by the old group.

In this problem:

- the young rank Z first,
- the middle-aged rank X first,
- the old rank Y first.

Hence, compared with standard Borda voting:

- Z should gain because the young are heavily up-weighted,
- X should also gain because the middle-aged are up-weighted,
- Y is likely to lose relative ground because the old group is not up-weighted.

So, *in general*, the modified rule tilts the social ranking in favour of the policies preferred by the young and middle-aged, especially the young.

- (b) Yes, the modified Borda voting rule satisfies the *Pareto condition*.

Recall the Pareto condition: if *every* individual strictly prefers policy A to policy B , then the social ranking must also place A above B .

Under Borda scoring with three alternatives, each voter gives more points to a higher-ranked alternative than to a lower-ranked one. So if every voter ranks A above B , then for every voter,

$$\text{score of } A > \text{score of } B.$$

Under the modified rule, these individual scores are multiplied by positive weights (2, 1.5, and 1). Since all these weights are strictly positive, the inequality is preserved:

$$\text{weighted score of } A > \text{weighted score of } B$$

for every voter.

Summing across all voters, the total weighted Borda score of A must exceed that of B . Therefore the social ranking also places A above B .

Hence:

The modified Borda rule satisfies the Pareto condition.

- (c) We now compute the winning policy under the two voting rules.

(i) Standard Borda voting

Using the standard 2, 1, 0 scoring rule:

Group	First	Second	Third
Young (300)	Z	Y	X
Middle Aged (300)	X	Z	Y
Old (400)	Y	X	Z

Compute total scores:

$$\begin{aligned} \text{Score}(X) &= 300(0) + 300(2) + 400(1) \\ &= 0 + 600 + 400 \\ &= 1000, \end{aligned}$$

$$\begin{aligned} \text{Score}(Y) &= 300(1) + 300(0) + 400(2) \\ &= 300 + 0 + 800 \\ &= 1100, \end{aligned}$$

$$\begin{aligned} \text{Score}(Z) &= 300(2) + 300(1) + 400(0) \\ &= 600 + 300 + 0 \\ &= 900. \end{aligned}$$

Therefore the standard Borda social ranking is

$$Y \succ X \succ Z.$$

So the chosen policy is

$$\boxed{Y.}$$

(ii) Modified Borda voting

Under the modified rule:

- young scores are multiplied by 2,
- middle-aged scores are multiplied by 1.5,
- old scores are unchanged.

So the effective contributions are:

Young contribution: $Z : 300(2)(2) = 1200, \quad Y : 300(1)(2) = 600, \quad X : 300(0)(2) = 0.$

Middle-aged contribution: $X : 300(2)(1.5) = 900, \quad Z : 300(1)(1.5) = 450,$
 $Y : 300(0)(1.5) = 0.$

Old contribution: $Y : 400(2) = 800, \quad X : 400(1) = 400, \quad Z : 400(0) = 0.$

Summing up:

$$\text{Modified Score}(X) = 0 + 900 + 400 = 1300,$$

$$\text{Modified Score}(Y) = 600 + 0 + 800 = 1400,$$

$$\text{Modified Score}(Z) = 1200 + 450 + 0 = 1650.$$

Therefore the modified Borda social ranking is

$$Z \succ Y \succ X.$$

So the chosen policy is

$$\boxed{Z.}$$

Hence:

Standard Borda chooses Y,	Modified Borda chooses Z.
---------------------------	---------------------------

- (d) A reasonable argument in favour of the modified Borda rule is that environmental policy has *long-term consequences*, so it may be appropriate to give greater weight to groups who will live with those consequences for longer.

In this scenario:

- young consumers are likely to be affected by climate and environmental outcomes over a longer remaining lifetime,
- middle-aged consumers also have substantial future exposure, and may additionally care about their children,
- old consumers may be affected for a shorter period.

Under standard Borda voting, each person has the same weight, regardless of how long they will experience the effects of the policy. The modified rule instead places more emphasis on the preferences of those with a longer future stake in the consequences of environmental policy.

So one may argue that the modified rule is more “reasonable” because it better reflects:

- intergenerational fairness,
- the long-run nature of environmental policy,
- the idea that those more affected by future outcomes should have greater influence.

However, this is a normative judgment rather than a purely mathematical one. A criticism is that it violates the principle of *equal political weight* or “one person, one vote.” So whether it is more reasonable depends on the ethical criterion being used.

A balanced conclusion is therefore:

The modified rule may be more reasonable if voting power should reflect long-run stakes in environmental outcomes, but less reasonable if every voter should have equal weight.

Question 2

Solution

The utility possibility frontier (UPF) is given by

$$U_I + U_J^2 = 10.$$

We assume utilities are non-negative, so the relevant part of the frontier is

$$U_I \geq 0, \quad U_J \geq 0.$$

- (a) The *utility possibility frontier* is the boundary of all utility pairs (U_I, U_J) that can be achieved in the economy when resources are allocated efficiently.

In other words:

- the set of feasible utility allocations gives all utility pairs the economy can generate,
- the *frontier* consists of the Pareto efficient utility pairs,
- at a point on the frontier, one person's utility can be increased only by decreasing the other person's utility.

Since

$$U_I + U_J^2 = 10,$$

we can write the frontier as

$$U_I = 10 - U_J^2$$

or equivalently

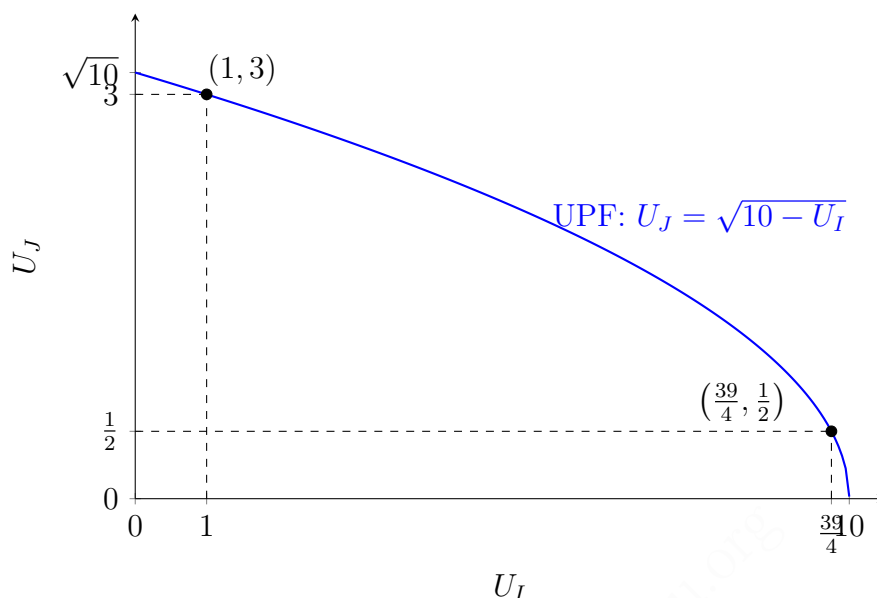
$$U_J = \sqrt{10 - U_I}.$$

The intercepts are:

$$U_J = 0 \implies U_I = 10, \quad U_I = 0 \implies U_J = \sqrt{10}.$$

So the UPF is downward sloping and bowed toward the origin in the (U_I, U_J) -diagram.

A suitable graph is:



The feasible utility set is the region on and below this frontier:

$$U_I + U_J^2 \leq 10.$$

(b) A utilitarian social planner maximizes the sum of utilities:

$$W(U_I, U_J) = U_I + U_J.$$

Since on the frontier

$$U_I = 10 - U_J^2,$$

substituting into the welfare function gives

$$W(U_J) = 10 - U_J^2 + U_J.$$

Differentiate:

$$\frac{dW}{dU_J} = -2U_J + 1.$$

Setting the first-order condition equal to zero,

$$-2U_J + 1 = 0 \implies U_J = \frac{1}{2}.$$

Since

$$\frac{d^2W}{dU_J^2} = -2 < 0,$$

this is indeed the welfare-maximizing point.

Then

$$U_I = 10 - \left(\frac{1}{2}\right)^2 = 10 - \frac{1}{4} = \frac{39}{4}.$$

Therefore the utilitarian planner's ideal allocation is

$$U_I^* = \frac{39}{4}, \quad U_J^* = \frac{1}{2}.$$

Graphically, this is the point where the highest attainable social indifference line

$$U_I + U_J = \bar{W}$$

is tangent to the utility possibility frontier.

(c) A weighted-utilitarian social welfare function has the form

$$W(U_I, U_J) = \alpha U_I + \beta U_J,$$

where $\alpha, \beta > 0$.

We are told that the planner's ideal allocation is

$$U_I = 1, \quad U_J = 3.$$

First check that this point lies on the utility possibility frontier:

$$1 + 3^2 = 1 + 9 = 10.$$

So it is feasible and efficient.

To find the weights, maximize

$$W = \alpha U_I + \beta U_J$$

subject to

$$U_I = 10 - U_J^2.$$

Substituting:

$$W(U_J) = \alpha(10 - U_J^2) + \beta U_J.$$

Differentiate:

$$\frac{dW}{dU_J} = -2\alpha U_J + \beta.$$

Setting the first-order condition equal to zero,

$$-2\alpha U_J + \beta = 0.$$

Since the optimum is required to occur at $U_J = 3$,

$$-2\alpha(3) + \beta = 0 \implies \beta = 6\alpha.$$

So the weights must satisfy

$$\frac{\beta}{\alpha} = 6.$$

One convenient choice is

$$\alpha = 1, \quad \beta = 6.$$

Hence an appropriate social welfare function is

$$W(U_I, U_J) = U_I + 6U_J.$$

Any positive scalar multiple of this is equivalent, for example

$$2U_I + 12U_J, \quad \frac{1}{6}U_I + U_J,$$

and so on.

Graphically, the planner's social indifference curves are straight lines

$$U_I + 6U_J = \bar{W}$$

or equivalently

$$U_J = \frac{\bar{W}}{6} - \frac{1}{6}U_I,$$

with slope

$$-\frac{1}{6}.$$

This matches the slope of the utility possibility frontier at $(1, 3)$, confirming tangency there.

Therefore:

$$\text{An appropriate weighted-utilitarian SWF is } W(U_I, U_J) = U_I + 6U_J.$$

- (d) Not every point on the utility possibility frontier will necessarily be achievable as a competitive equilibrium *from a given initial endowment*.

The important ideas are:

- Every competitive equilibrium allocation is Pareto efficient under the standard assumptions. This is the *First Welfare Theorem*.
- But the converse is more subtle. A Pareto efficient allocation need not arise as a competitive equilibrium for the *current* initial endowment.
- However, under the standard convexity assumptions, every Pareto efficient allocation can be supported as a competitive equilibrium for *some* redistribution of initial endowments. This is the *Second Welfare Theorem*.

Therefore the correct discussion is:

- If the initial endowment is fixed, then *not every* point on the UPF will generally be obtained as a competitive equilibrium.

- If the planner is allowed to redistribute endowments appropriately first, and if the economy satisfies the standard assumptions (convex preferences, local non-satiation, etc.), then every point on the UPF can be decentralized as a competitive equilibrium.

So the best conclusion is:

Not necessarily for a fixed endowment, but under the standard assumptions every point on the UPF can be supported as a competitive equilibrium after a suitable redistribution of endowments.

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Sample Question 1: Weighted Borda Voting

Problem

There are three policies X, Y, Z , and three preference types:

Young: $Z \succ Y \succ X$, Middle Aged: $X \succ Z \succ Y$, Old: $Y \succ X \succ Z$.

The numbers of voters are:

300 young, 300 middle aged, 400 old.

Under *standard Borda voting* with three alternatives, each voter awards:

1st place = 2 points, 2nd place = 1 point, 3rd place = 0 points.

Under the *modified Borda voting* rule, the same ranking scores apply, except:

- young voters' points are multiplied by 2,
 - middle-aged voters' points are multiplied by 1.5,
 - old voters' points remain unchanged.
- (a) In general, would the modified Borda voting rule make the social decision tend to align more with the old, the middle-aged, or the young? Explain briefly.
- (b) Does the modified Borda voting rule satisfy the Pareto condition? Explain briefly.
- (c) Which policy would be chosen under the standard Borda voting rule, and which policy would be chosen under the modified Borda voting rule?
- (d) Do you think the modified Borda voting rule is more reasonable in this environmental policy context? Briefly explain.

Solution

Method 1: Direct weighted-score computation

- (a) In general, the modified Borda rule gives more weight to the preferences of young and middle-aged consumers relative to old consumers.

Under standard Borda, each person's ranking contributes equally. Under the modified rule:

$$\text{Young weight} = 2, \quad \text{Middle-aged weight} = 1.5, \quad \text{Old weight} = 1.$$

So the social ranking will tend to shift toward alternatives that are ranked highly by the young and middle-aged groups, and away from alternatives that are mainly supported by the old group.

In this problem:

- the young rank Z first,
- the middle-aged rank X first,
- the old rank Y first.

Hence, compared with standard Borda voting:

- Z should gain because the young are heavily up-weighted,
- X should also gain because the middle-aged are up-weighted,
- Y is likely to lose relative ground because the old group is not up-weighted.

Therefore

The modified rule tends to align more with the young and middle-aged, especially the young.

- (b) Yes, the modified Borda voting rule satisfies the Pareto condition.

If every individual strictly prefers policy A to policy B , then under ordinary Borda scoring each person assigns strictly more points to A than to B . Under the modified rule, these scores are multiplied by strictly positive weights 2, 1.5, and 1, so the strict inequality is preserved for every voter.

Summing across all voters, the total weighted Borda score of A must exceed that of B . Therefore the social ranking also places A above B .

Hence

the modified Borda rule satisfies the Pareto condition.

- (c) **Standard Borda voting.**

The three groups rank:

$$\text{Young: } Z \succ Y \succ X, \quad \text{Middle Aged: } X \succ Z \succ Y, \quad \text{Old: } Y \succ X \succ Z.$$

Using the standard 2, 1, 0 scoring rule:

$$\text{Score}(X) = 300(0) + 300(2) + 400(1) = 1000,$$

$$\text{Score}(Y) = 300(1) + 300(0) + 400(2) = 1100,$$

$$\text{Score}(Z) = 300(2) + 300(1) + 400(0) = 900.$$

Therefore the standard Borda social ranking is

$$Y \succ X \succ Z,$$

so the chosen policy is

$$\boxed{Y.}$$

Modified Borda voting.

Under the modified rule:

- young scores are multiplied by 2,
- middle-aged scores are multiplied by 1.5,
- old scores are unchanged.

So the weighted contributions are:

$$\begin{aligned} \text{Modified Score}(X) &= 300(0)(2) + 300(2)(1.5) + 400(1)(1) \\ &= 0 + 900 + 400 = 1300, \end{aligned}$$

$$\begin{aligned} \text{Modified Score}(Y) &= 300(1)(2) + 300(0)(1.5) + 400(2)(1) \\ &= 600 + 0 + 800 = 1400, \end{aligned}$$

$$\begin{aligned} \text{Modified Score}(Z) &= 300(2)(2) + 300(1)(1.5) + 400(0)(1) \\ &= 1200 + 450 + 0 = 1650. \end{aligned}$$

Therefore the modified Borda social ranking is

$$Z \succ Y \succ X,$$

so the chosen policy is

$$\boxed{Z.}$$

Hence

$$\boxed{\text{Standard Borda chooses } Y, \quad \text{Modified Borda chooses } Z.}$$

- (d) A reasonable argument in favour of the modified Borda rule is that environmental policy has long-term consequences, so it may be appropriate to give greater weight to groups who will live with those consequences for longer.

In this scenario:

- young consumers are likely to be affected by environmental outcomes over a longer remaining lifetime,

- middle-aged consumers also have substantial future exposure,
- old consumers may be affected for a shorter period.

Under standard Borda voting, each person has the same weight regardless of how long they will experience the effects of the policy. The modified rule instead places more emphasis on the preferences of those with a longer future stake.

A criticism, however, is that it violates the principle of equal political weight or “one person, one vote.” So whether it is more reasonable depends on the ethical principle being used.

A balanced conclusion is:

The modified rule may be more reasonable if voting power should reflect long-run stakes in environmental outcomes, but less reasonable if every voter should have equal weight.

Method 2: Linear-functional / vector method

- (a) Write the standard Borda score vector for (1st, 2nd, 3rd) as

$$s = (2, 1, 0).$$

The modified rule is simply a reweighting of group contributions by

$$w = (2, 1.5, 1).$$

Since the young and middle-aged groups are upweighted relative to the old group, the induced social ranking shifts toward the alternatives they rank highly. Since the young place Z first and the middle-aged place X first, while the old place Y first, the modified rule moves weight away from Y and toward Z and X , especially Z .

- (b) The Pareto condition follows immediately from positivity of the weights. If every voter ranks A above B , then every group contributes a strictly larger Borda score to A than to B . Multiplying by positive weights preserves the inequality, so total weighted score still ranks A above B .
- (c) Let the score vector for each policy under the three groups be:

$$X = (0, 2, 1), \quad Y = (1, 0, 2), \quad Z = (2, 1, 0).$$

Under standard Borda, with population vector

$$n = (300, 300, 400),$$

the total scores are proportional to

$$\langle n, X \rangle = 1000, \quad \langle n, Y \rangle = 1100, \quad \langle n, Z \rangle = 900.$$

Hence

$$Y \succ X \succ Z.$$

Under modified Borda, the effective population vector is

$$n' = (600, 450, 400),$$

so

$$\langle n', X \rangle = 1300, \quad \langle n', Y \rangle = 1400, \quad \langle n', Z \rangle = 1650.$$

Hence

$$Z \succ Y \succ X.$$

Therefore standard Borda chooses Y and modified Borda chooses Z .

- (d) In this representation, the modified rule changes the social ranking by changing the weight vector. So the normative question is whether one regards this change of weight vector as ethically justified. If voting weight should reflect future stake, the modification is defensible; if all persons should count equally, it is not.

Method 3: Pairwise score-difference method

- (a) Under the modified rule, the relative influence of the young and middle-aged groups rises, while the old group keeps weight 1. Therefore alternatives favoured by the young and middle-aged gain relative to alternatives favoured mainly by the old.
- (b) If everyone ranks A above B , then each individual contributes a strictly higher Borda score to A than to B . Positive reweighting preserves this group-by-group advantage, so the aggregate ranking still has A above B . Thus Pareto holds.
- (c) Instead of computing all totals, compare relevant score differences.

Under standard Borda:

$$\begin{aligned}\text{Score}(Y) - \text{Score}(X) &= 300(1 - 0) + 300(0 - 2) + 400(2 - 1) \\ &= 300 - 600 + 400 = 100 > 0,\end{aligned}$$

so $Y \succ X$.

Also,

$$\begin{aligned}\text{Score}(X) - \text{Score}(Z) &= 300(0 - 2) + 300(2 - 1) + 400(1 - 0) \\ &= -600 + 300 + 400 = 100 > 0,\end{aligned}$$

so $X \succ Z$. Hence

$$\boxed{Y \succ X \succ Z.}$$

Under modified Borda:

$$\begin{aligned}\text{Score}(Z) - \text{Score}(Y) &= 300 \cdot 2(2 - 1) + 300 \cdot 1.5(1 - 0) + 400 \cdot 1(0 - 2) \\ &= 600 + 450 - 800 = 250 > 0,\end{aligned}$$

so $Z \succ Y$.

Next,

$$\begin{aligned}\text{Score}(Y) - \text{Score}(X) &= 300 \cdot 2(1 - 0) + 300 \cdot 1.5(0 - 2) + 400 \cdot 1(2 - 1) \\ &= 600 - 900 + 400 = 100 > 0,\end{aligned}$$

so $Y \succ X$. Hence

$$\boxed{Z \succ Y \succ X.}$$

- (d) Since the modified rule makes Z overtake Y precisely because of the heavier weight placed on the young and middle-aged, the reasonableness of the rule depends on whether that reweighting is normatively acceptable.

Marking scheme (20 marks)

- (a) **(4 marks)** Comparative effect of the modified weights
- 2 marks: Correct identification that the modified rule shifts influence toward the young and middle-aged, especially the young.
 - 2 marks: Clear explanation using the weight changes 2, 1.5, 1 and the ranking pattern of the three groups.
- (b) **(4 marks)** Pareto condition
- 2 marks: Correct conclusion that the modified Borda rule satisfies the Pareto condition.
 - 2 marks: Correct justification that positive weighting preserves the strict score advantage of a unanimously preferred alternative.
- (c) **(8 marks)** Policy choice under standard and modified Borda
- 3 marks: Correct standard Borda score calculation or an equivalent correct derivation.
 - 3 marks: Correct modified Borda score calculation or an equivalent correct derivation.
 - 2 marks: Correct final conclusion that standard Borda chooses Y while modified Borda chooses Z .
- (d) **(4 marks)** Reasonableness in context
- 2 marks: A coherent argument in favour of the modified rule based on long-run stake / intergenerational exposure.
 - 2 marks: Recognition of the counterargument concerning equal political weight, together with a balanced conclusion.

Sample Question 2: Utility Possibility Frontier and Welfare Optimisation

Problem

The utility possibility frontier (UPF) is given by

$$U_I + U_J^2 = 10.$$

We assume utilities are non-negative, so the relevant part of the frontier is

$$U_I \geq 0, \quad U_J \geq 0.$$

- (a) Explain what the utility possibility frontier means, and sketch it in the (U_I, U_J) -plane.
- (b) If a utilitarian social planner maximizes

$$W(U_I, U_J) = U_I + U_J,$$

what is the planner's ideal allocation?

- (c) If a weighted-utilitarian social welfare function is used, and the planner's ideal allocation is

$$U_I = 1, \quad U_J = 3,$$

what weighted-utilitarian social welfare function would support this point?

- (d) Is every point on the utility possibility frontier necessarily obtainable as a competitive equilibrium? Briefly explain.

Solution

Method 1: Direct substitution and tangency

- (a) The *utility possibility frontier* is the boundary of all utility pairs (U_I, U_J) that can be achieved in the economy when resources are allocated efficiently.

In other words:

- the feasible utility set consists of all utility pairs the economy can generate,
- the *frontier* consists of the Pareto efficient utility pairs,
- at a point on the frontier, one person's utility can be increased only by decreasing the other person's utility.

Since

$$U_I + U_J^2 = 10,$$

we can write the frontier as

$$U_I = 10 - U_J^2$$

or equivalently

$$U_J = \sqrt{10 - U_I}.$$

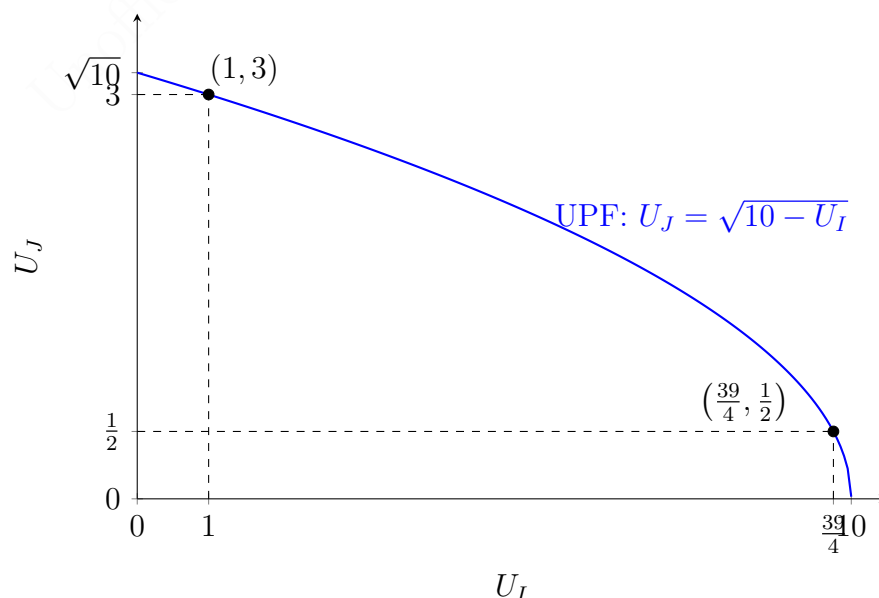
The intercepts are:

$$U_J = 0 \implies U_I = 10, \quad U_I = 0 \implies U_J = \sqrt{10}.$$

So the UPF is downward sloping and bowed toward the origin in the (U_I, U_J) -diagram, and the feasible set is

$$U_I + U_J^2 \leq 10.$$

A suitable graph is:



(b) A utilitarian social planner maximizes

$$W(U_I, U_J) = U_I + U_J.$$

Since on the frontier

$$U_I = 10 - U_J^2,$$

substitute into welfare:

$$W(U_J) = 10 - U_J^2 + U_J.$$

Differentiate:

$$\frac{dW}{dU_J} = -2U_J + 1.$$

Set equal to zero:

$$-2U_J + 1 = 0 \implies U_J = \frac{1}{2}.$$

Then

$$U_I = 10 - \left(\frac{1}{2}\right)^2 = 10 - \frac{1}{4} = \frac{39}{4}.$$

Therefore the utilitarian optimum is

$$\boxed{U_I^* = \frac{39}{4}, \quad U_J^* = \frac{1}{2}.}$$

(c) A weighted-utilitarian social welfare function has the form

$$W(U_I, U_J) = \alpha U_I + \beta U_J, \quad \alpha, \beta > 0.$$

We are told that the ideal allocation is

$$U_I = 1, \quad U_J = 3.$$

First check feasibility:

$$1 + 3^2 = 10.$$

So the point lies on the frontier.

Substitute $U_I = 10 - U_J^2$ into welfare:

$$W(U_J) = \alpha(10 - U_J^2) + \beta U_J.$$

Differentiate:

$$\frac{dW}{dU_J} = -2\alpha U_J + \beta.$$

Set equal to zero at $U_J = 3$:

$$-2\alpha(3) + \beta = 0 \implies \beta = 6\alpha.$$

So the required weight ratio is

$$\boxed{\frac{\beta}{\alpha} = 6.}$$

One valid supporting welfare function is

$$\boxed{W(U_I, U_J) = U_I + 6U_J.}$$

- (d) Not every point on the utility possibility frontier is necessarily obtainable as a competitive equilibrium from a *given* initial endowment.

The relevant economic principles are:

- The *First Welfare Theorem*: every competitive equilibrium allocation is Pareto efficient under the usual assumptions.
- The *Second Welfare Theorem*: under suitable convexity assumptions, every Pareto efficient allocation can be decentralised as a competitive equilibrium for *some* suitable redistribution of endowments.

Therefore:

Not necessarily for a fixed endowment, but under the standard assumptions every point on the UPF can be supported as a competitive equilibrium after a suitable redistribution of endowments.

Method 2: Lagrangian method

- (a) The UPF is the boundary of the feasible utility set

$$\{(U_I, U_J) \in \mathbb{R}_+^2 : U_I + U_J^2 \leq 10\}.$$

The frontier itself is

$$U_I + U_J^2 = 10,$$

with intercepts $(10, 0)$ and $(0, \sqrt{10})$.

- (b) Maximize

$$U_I + U_J$$

subject to

$$U_I + U_J^2 = 10.$$

The Lagrangian is

$$\mathcal{L} = U_I + U_J + \lambda(10 - U_I - U_J^2).$$

First-order conditions:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial U_I} &= 1 - \lambda = 0, \\ \frac{\partial \mathcal{L}}{\partial U_J} &= 1 - 2\lambda U_J = 0.\end{aligned}$$

From the first equation,

$$\lambda = 1.$$

Substitute into the second:

$$1 - 2U_J = 0 \implies U_J = \frac{1}{2}.$$

Then

$$U_I = 10 - \left(\frac{1}{2}\right)^2 = \frac{39}{4}.$$

Hence

$$\boxed{(U_I^*, U_J^*) = \left(\frac{39}{4}, \frac{1}{2}\right)}.$$

(c) Now maximize

$$\alpha U_I + \beta U_J$$

subject to

$$U_I + U_J^2 = 10.$$

The Lagrangian is

$$\mathcal{L} = \alpha U_I + \beta U_J + \lambda(10 - U_I - U_J^2).$$

FOCs:

$$\begin{aligned}\alpha - \lambda &= 0, \\ \beta - 2\lambda U_J &= 0.\end{aligned}$$

Hence

$$\lambda = \alpha, \quad \beta = 2\alpha U_J.$$

At the required point $U_J = 3$,

$$\beta = 6\alpha.$$

Therefore

$$\boxed{\frac{\beta}{\alpha} = 6},$$

and one valid welfare function is

$$\boxed{W(U_I, U_J) = U_I + 6U_J}.$$

- (d) The Lagrangian analysis solves a planner's problem, not necessarily a competitive-equilibrium problem for a fixed endowment. Competitive equilibrium requires decentralisation from a specific wealth distribution. Thus a point may be planner-efficient without being a competitive equilibrium for the original endowment vector. With appropriate redistribution and the standard assumptions, however, the Second Welfare Theorem allows decentralisation of every Pareto efficient point.

Method 3: Convex-analysis / supporting-hyperplane method

- (a) Define the feasible utility set

$$\mathcal{U} = \{(U_I, U_J) \in \mathbb{R}_+^2 : U_I + U_J^2 \leq 10\}.$$

Since U_J^2 is convex, the set $\mathcal{U}$ is convex. Its boundary

$$U_I + U_J^2 = 10$$

is the utility possibility frontier.

- (b) The utilitarian planner maximizes the linear functional

$$(U_I, U_J) \mapsto U_I + U_J$$

over the convex set $\mathcal{U}$. Therefore the optimum is the point on the boundary where the supporting line with normal vector $(1, 1)$ touches the frontier.

The frontier is the zero set of

$$F(U_I, U_J) = U_I + U_J^2 - 10.$$

Its normal vector is

$$\nabla F = (1, 2U_J).$$

At a supporting point for utilitarian welfare, the normal vector to the frontier must be parallel to the normal vector of the welfare level sets, namely $(1, 1)$. So

$$(1, 2U_J) \parallel (1, 1).$$

Hence

$$2U_J = 1 \implies U_J = \frac{1}{2}.$$

Therefore

$$U_I = 10 - \frac{1}{4} = \frac{39}{4}.$$

Thus

$$(U_I^*, U_J^*) = \left(\frac{39}{4}, \frac{1}{2} \right).$$

- (c) For weighted-utilitarian welfare

$$W(U_I, U_J) = \alpha U_I + \beta U_J,$$

the supporting hyperplanes have normal vector

$$(\alpha, \beta).$$

At $(1, 3)$, the frontier normal vector is

$$\nabla F(1, 3) = (1, 6).$$

Therefore the supporting welfare weights must satisfy

$$(\alpha, \beta) \parallel (1, 6).$$

So

$$\boxed{\frac{\beta}{\alpha} = 6,}$$

and one valid welfare function is

$$\boxed{W(U_I, U_J) = U_I + 6U_J.}$$

- (d) The convex-analysis perspective makes the decentralisation issue transparent: competitive equilibria correspond to support by price hyperplanes from a given endowment, whereas Pareto efficient points are merely supportable boundary points of the feasible utility set. Thus supportability by a planner's welfare functional is weaker than supportability by competitive equilibrium from a fixed initial allocation.

Method 4: Duality / planner-support method

- (a) The UPF is the *primal* efficient boundary

$$U_I + U_J^2 = 10$$

of the feasible utility set. The planner's welfare problem is the corresponding *dual support problem*: choose a linear welfare functional whose highest attainable indifference line touches the frontier.

- (b) For utilitarian welfare

$$W(U_I, U_J) = U_I + U_J,$$

the dual problem is

$$\max\{U_I + U_J : U_I + U_J^2 \leq 10\}.$$

The support line has slope -1 . The frontier has slope

$$\frac{dU_I}{dU_J} = -2U_J.$$

Dual optimality requires equality of slopes:

$$-2U_J = -1.$$

Hence

$$U_J = \frac{1}{2}, \quad U_I = \frac{39}{4}.$$

Therefore

$$(U_I^*, U_J^*) = \left(\frac{39}{4}, \frac{1}{2}\right).$$

- (c) If a point $(1, 3)$ is to be planner-optimal under a weighted-utilitarian functional, then the dual support line through that point must have the same slope as the frontier there. Since the frontier slope at $U_J = 3$ is

$$-2U_J = -6,$$

the welfare line

$$\alpha U_I + \beta U_J = \bar{W}$$

must have slope

$$-\frac{\alpha}{\beta} = -\frac{1}{6} \implies \frac{\beta}{\alpha} = 6.$$

Thus

$$W(U_I, U_J) = U_I + 6U_J$$

is a valid supporting welfare function.

- (d) Duality also clarifies the equilibrium question. A planner's welfare function can support any efficient point by choosing an appropriate linear objective. Competitive equilibrium support is more restrictive: the supporting prices and wealth distribution must arise from market decentralisation. Therefore not every efficient point is a competitive equilibrium for a given initial endowment, although under the standard assumptions each efficient point can be decentralised after suitable redistribution.

Marking scheme (20 marks)**(a) (5 marks)** Meaning and graph of the UPF

- 2 marks: Correct explanation that the UPF is the set of Pareto efficient feasible utility allocations.
- 1 mark: Correct intercepts $(10, 0)$ and $(0, \sqrt{10})$ or an equivalent correct characterisation.
- 2 marks: Correct sketch and correct description of the feasible set $U_I + U_J^2 \leq 10$.

(b) (5 marks) Utilitarian optimum

- 2 marks: Correct optimisation setup.
- 2 marks: Correct solution $(\frac{39}{4}, \frac{1}{2})$.
- 1 mark: Correct justification that this is the maximiser.

(c) (5 marks) Weighted-utilitarian support

- 2 marks: Correct tangency / first-order condition linking weights to the slope at $(1, 3)$.
- 2 marks: Correct weight ratio $\beta/\alpha = 6$.
- 1 mark: A correct supporting welfare function, e.g. $W(U_I, U_J) = U_I + 6U_J$.

(d) (5 marks) Competitive-equilibrium interpretation

- 2 marks: Correct distinction between fixed endowment and redistributed endowment.
- 2 marks: Correct use of the First and / or Second Welfare Theorem.
- 1 mark: Clear final conclusion.